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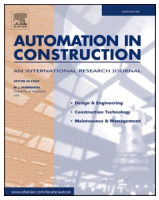

Digital twin model for analyzing deformation and seepage in high earth-rock dams

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ABSTRACT

Digital twin technology is vital in hydraulic engineering for real-time visualization, performance analysis, and risk management of water infrastructure systems. This paper proposes a digital twin model for deformation and seepage analysis of high earth-rock dams, integrating deep learning with the Finite Element Method (FEM). Key contributions include a sensor-based monitoring point model with time-variant update and extrapolation capabilities and a point-to-domain model that achieves full-domain monitoring predictions from point-level monitoring by learning the node relationships generated by FEM using neural networks and dynamic monitoring loss functions. Applied to a 186-m dam, the model achieves an average error of 3.17 %, improving deformation prediction accuracy by 19.44 % and simulation accuracy by 64.42 %. This approach facilitates real-time monitoring, predictive analysis, and early warnings, making it a powerful tool for hydraulic engineering safety. Future work will focus on exploring three-dimensional high-precision modeling and advancing data fusion techniques.

1. Introduction

Earth-rock dams are widely utilized in global water conservancy projects due to their low cost, easy access to local materials, adaptability to diverse topographical and geological conditions, simplicity of construction, and excellent seismic performance [1,2]. Driven by the rapid development of hydropower in China, numerous high earth-rock dams are already in the design and construction stages, such as the Gushui Dam (305 m), the Shuangjiangkou Dam (315 m), the Rumei Dam (315 m), and the Lianghekou Dam (295 m) [3,4]. These large-scale watershed-controlling high earth-rock dams are often located in mountainous canyon regions, where they face complex environmental conditions, such as large water level fluctuations, significant stresses, and thick overburden layers [5,6]. Real-time monitoring and safety analysis of deformation and seepage for high earth-rock dams in these areas are crucial for preventing structural failure and ensuring safe operation.

Traditionally, real-time monitoring of the structural health of high earth-rock dams has relied on data-driven models based on sensor monitoring points, such as the Hydrostatic-Season-Time (HST) model, which is limited by the linear assumptions and fixed parameters, reducing accuracy in capturing complex, non-linear behaviors and dynamic changes in dam safety [7–10]. With advancements in machine learning, particularly deep learning, more powerful models with superior modeling and feature extraction capabilities have been introduced for dam safety monitoring [11–13]. However, capturing the complex relationships between dam deformation and various environmental factors, as well as modeling temporal features, remains a challenge for both the statistical models and shallow machine-learning approaches [14]. Thus, deep learning models, represented by neural networks, have emerged to effectively address issues such as multi-feature relationships and multi-point correlations in dam safety monitoring. For example, Recurrent Neural Networks (RNNs) like Long Short-Term Memory (LSTM) [15,16] and Gated Recurrent Units (GRU) [17] have been employed to capture the long-term futures of dams. Graph Neural Networks (GNNs) have shown particular promise in capturing the complex spatial dependencies of sensor location graphs [18], with models such as Graph Convolutional Network (GCN) being applied by researchers to develop multi-point monitoring models for dam structural health

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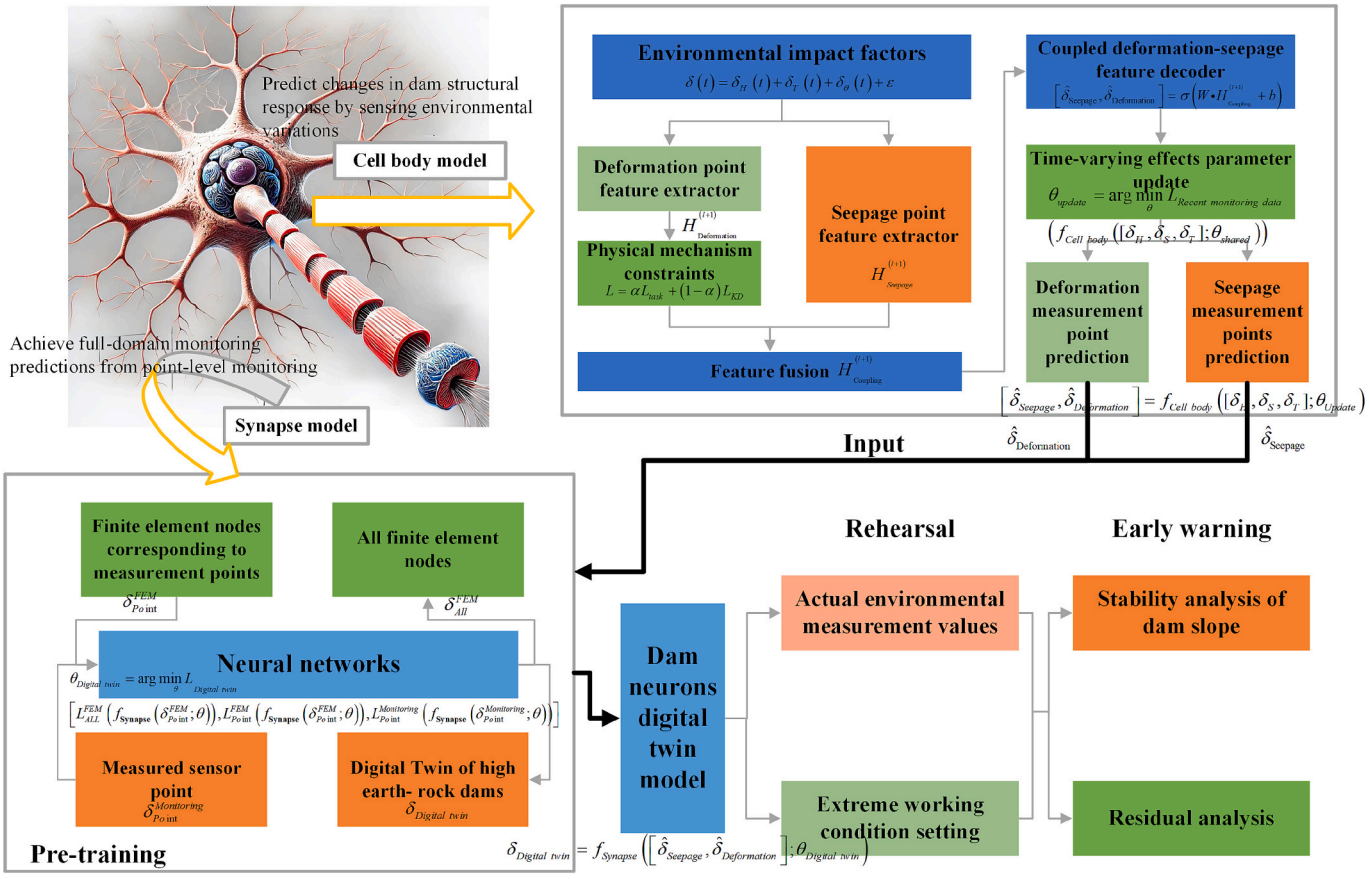


Fig. 1. Rationale for the proposed model.

[19,20]. Additionally, attention mechanisms, known for their powerful information processing capabilities [21], have facilitated the use of Transformer-based models [22] in dam safety monitoring applications.

Although deep learning models exhibit exceptional nonlinear modeling capabilities for high earth-rock dam structures, their extrapolation performance under extreme water level conditions such as dead water level and check flood water level remains limited. Pre-training methods based on transfer learning show promise in providing prior knowledge to deep learning-based monitoring models for extreme and previously unobserved conditions. Transfer learning leverages knowledge learned in one domain to improve learning tasks in the same or different domains, thereby enhancing model generalization [23]. In the field of dam safety monitoring, transfer learning methods based on pre-training and fine-tuning have been primarily employed to address challenges such as insufficient labeled crack images of dams [22,24] and high earth-rock particle fragmentation [2], which often lead to poor model generalization. By utilizing prior task knowledge, transfer learning not only enhances prediction accuracy but also improves overall efficiency [25], leading to a significant improvement in the reliability and accuracy of models used in dam safety monitoring [26].

The aforementioned dam monitoring model based on single or multiple sensor-based monitoring points is insufficient for assessing the structural health of high earth-rock dams in complex environments [27], as an evaluation of the overall structural condition of the project may be indispensable. Alternatively, the digital twin model offers a promising solution [28]. It refers to the interaction between a virtual model and real-time data of physical entities, processes, or systems, establishing a digital, dynamically synchronized mapping relationship. This model can reflect the state, behavior, and changes of the physical object in real-time and, through simulation, prediction, and optimization functions, enable intelligent monitoring, analysis, and decision support for physical systems [29]. It includes three core components: the physical entity, its digital representation, and the data flow and feedback connecting them, supported by data sensing technologies, artificial intelligence capabilities, and high-speed networks [30]. However, the construction of existing digital twin models for dam structures primarily relies on numerical simulation frameworks based on Finite Element Model (FEM) [31,32], which often requires extensive material parameter calibration, structural simplifications, and iterative equation solving [33], limiting both timeliness and accuracy [34]. Additionally, some researchers have developed geographic information system-based geospatial digital twin models for dams and river systems, achieving high-precision digitalization of engineering structures [35], but these models cannot dynamically respond to environmental changes in real-time. In other fields, the development of digital twin models for small-scale physical systems is often based on numerical simulations, with machine learning models facilitating the connection between physical and digital environments [36–38]. Therefore, key obstacles include the construction of physical-digital models, the integration of sensor data, and the establishment of robust connections and feedback loops between the physical environment and the digital model. It reveals a critical gap in the availability of a reliable digital twin model for dam structural safety monitoring. Transitioning from traditional monitoring point models to digital twin models for dam structures remains a significant challenge that requires urgent investigation.

Therefore, this paper proposes a deformation-seepage digital twin model for high earth-rock dams, which integrates deep learning and FEM, for the first time. Firstly, a novel monitoring model with time-update and extrapolation capabilities under extreme water level conditions is constructed using GCN and transfer learning. Subsequently, by integrating LSTM with the FEM, monitoring point data from the monitoring point model can be mapped to full-domain monitoring

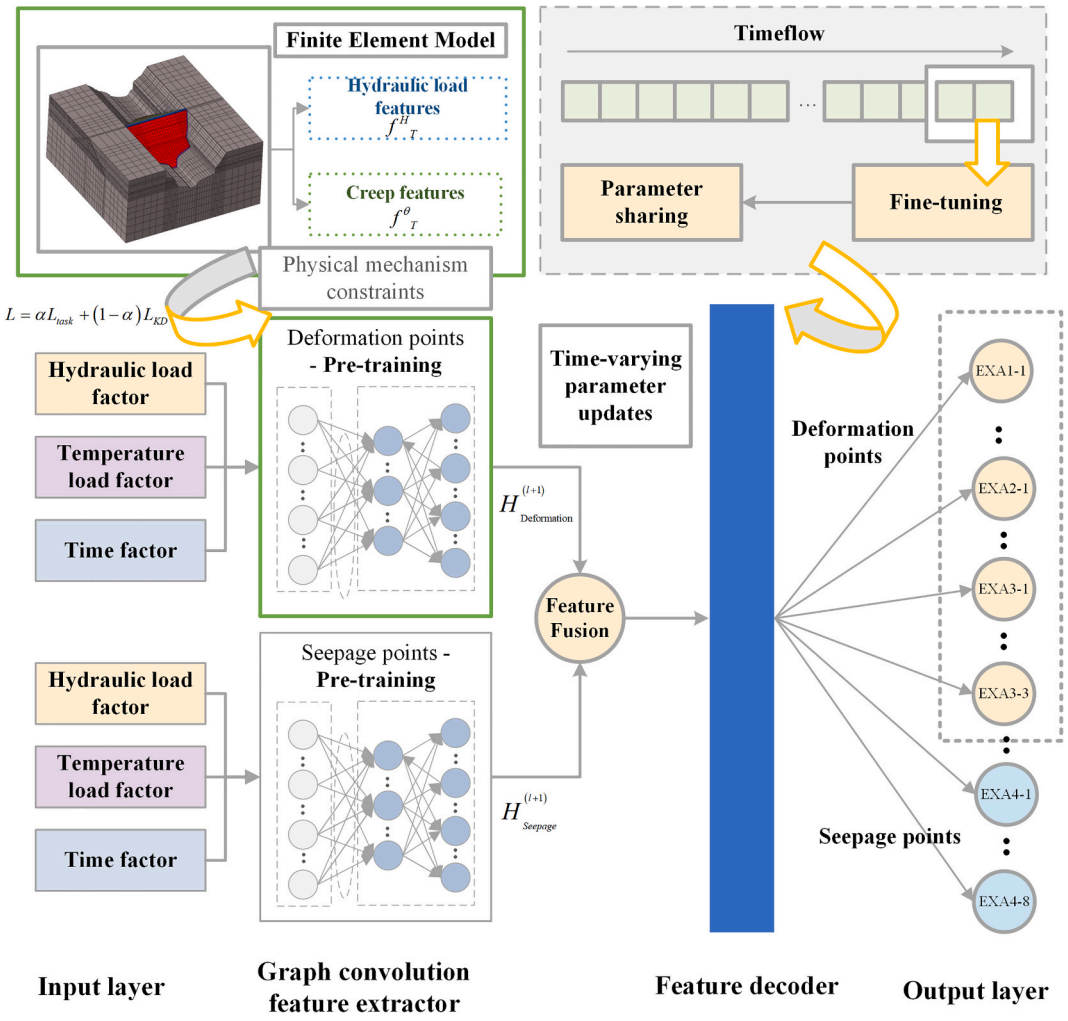


Fig. 2. Fundamentals of the cell body model.

information. Then, a dynamic loss function is established to effectively fuse sensor data with FEM outputs. The proposed digital twin model is applied to a 186-m earth-rock dam to verify its high accuracy and effectiveness. This paper represents the first attempt to apply digital twin technology to dam structural safety monitoring, marking a milestone in the field and offering a groundbreaking approach to enhancing dam safety and management.

2. Theory of monitoring traditional earth-rock dams

Whether employing traditional statistical models or machine learning models, most existing dam monitoring models are constructed based on the HST influence factor theory [39]. The HST model categorizes dam deformation into two types: reversible and irreversible. Reversible deformation is induced by fluctuations in external environmental loads, including changes in hydrostatic water level [41] and temperature [42]. In contrast, irreversible deformation is attributed to nonlinear effects such as creep, which can potentially compromise the structural integrity [40]. The formula for the HST model is typically expressed as follows [43]:

$$\delta(t) = \delta_H(t) + \delta_T(t) + \delta_\theta(t) + \varepsilon \tag{1}$$

Where $\delta(t)$ denotes the measured displacement and where $\delta_H(t)$, $\delta_T(t)$, and $\delta_\theta(t)$ represent the hydraulic, temperature, and aging components, respectively. ε denotes the regression residual.

3. Finite element method

The FEM, as one of the commonly used numerical analysis models for calculating stress and strain in complex nonlinear structures [44], is widely applied in modern dam engineering [45]. The fundamental concept of the finite element method involves dividing a complex structure into a finite number of non-overlapping elements that are interconnected [46]. For simulating the dams' instantaneous deformation, the Duncan-Chang *E*-B model is commonly used [47]. Creep calculations often rely on the empirical creep constitutive models [48]. In the analysis of seepage in earth-rock dams, it is generally assumed that the movement of seepage water conforms to Darcy's law. Based on this principle, seepage control equations are formulated [49].

4. Proposed model

Section 4 introduces the Dam Neurons Digital Twin Model, a digital twin model for high earth-rock dams that integrates deep learning, transfer learning, and FEM. The model dynamically predicts seepage and deformation states under varying environmental and operational conditions, enabling real-time monitoring, safety evaluation, and early warning.

4.1. General description

Establishing a digital representation of the physical model for high

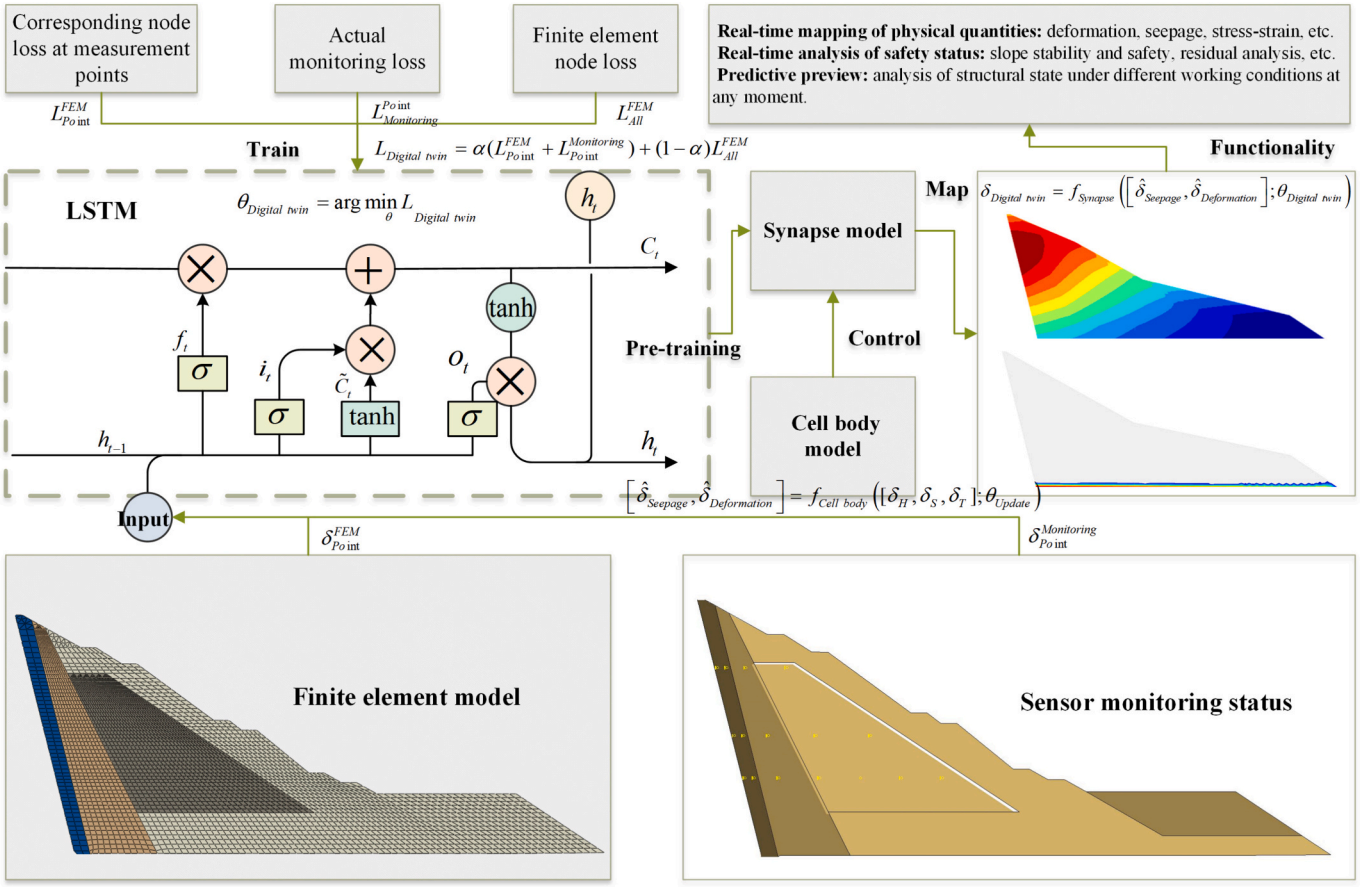


Fig. 3. Fundamentals of the synapse model.

earth-rock dams and integrating sensor monitoring information to connect feedback between the digital model and the real world presents a significant challenge. To provide a more economical and reliable monitoring solution for high earth-rock dam structures, this paper proposes a digital twin model, called the Dam Neurons Digital Twin Model, which leverages deep learning, transfer learning, and finite element techniques to dynamically reflect the seepage and deformation states of the dam. This model is capable of predicting the cross-sectional states of the dam under different water levels, temperatures, and operational conditions, while also providing real-time evaluation and early warning for seepage safety and slope stability. The implementation process is illustrated in Fig. 1. The approach begins by constructing a monitoring point model for monitoring seepage and deformation at sensor points, based on GCN, transfer learning, and physical feature constraints. This model enables the high earth-rock dam structure to sense and simulate the impact of environmental factors. Additionally, it possesses time-variant updating capabilities and extrapolates structural changes under extreme water level conditions. Next, a point-to-domain mapping model is developed to connect sensor points with surrounding nodes, utilizing LSTM and FEM. This model is built by learning the node relationships generated by FEM using LSTM and dynamic monitoring loss functions. The monitoring point model predicts structural response changes in the dam by sensing environmental variations. Its predicted values are then input into the point-to-domain mapping model to achieve full-domain monitoring predictions from point-level monitoring. To better illustrate this approach, inspired by the bionic concept of information transmission between neuronal cell bodies and synapses, the monitoring point model and the point-to-domain mapping model are referred to as the "cell body model" and "synapse model", respectively. Finally, the integration of the cell body model and dendrite model enables real-time perception, prediction, and early warning [50] of seepage and deformation in the dams, while further assessing structural safety through slope stability analysis. The main contributions of this paper are as follows:

- For the first time, a new digital twin model suitable for high earth-rock dams is proposed.
- A new integrated monitoring model for seepage and deformation sensor points with time-variant updating and extrapolation capabilities under extreme water level conditions is introduced.
- A model for jointly mapping numerical simulations and monitoring data to physical entities is proposed.
- The model presented in this paper is applied to a real-world high earth-rock dam project, with extensive accuracy and applicability verification conducted.

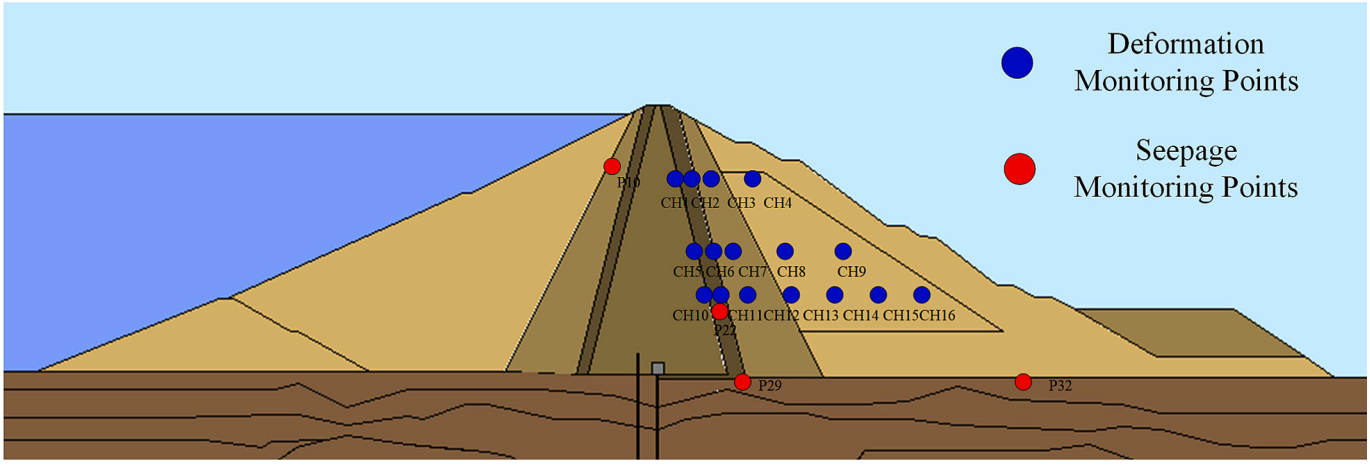
4.2. Cell body model

The principle of the cell body model is shown in Fig. 2. First, a feature extraction module based on GCN extracts the influence factor features. To couple the dams' seepage and deformation futures, separate feature extractors for seepage and deformation sensor points are constructed, and feature fusion is performed using pre-trained models. The hydraulic load features and soil-rock creep futures output from the FEM are used to apply physical constraints to the deformation sensor feature extractor, enhancing the model's ability to extrapolate structural responses under extreme high-water conditions. Finally, transfer learning is applied to update the time-variant effects of the feature decoder, mitigating performance degradation caused by issues such as particle fragmentation.

4.2.1. Input layer

Based on the HST model, sets of hydraulic load influence factors,

(a)



(b)

Fig. 4. Project overview. (a) Upstream view of PB Dam. (b) 210 Section deformation and seepage measurement point placement.

Table 1
PB dataset specifics table.

Data set	Length	Division ratio	Input dimension			Output points
			Hydraulic	Creep	Temperature	
Cell body model	4381	7:1:2	4	3	4	20
Synapse model-deformation	4381	7:1:2	Number of measurement points = 16			3238
Synapse model-seepage	4381	7:1:2	Number of measurement points = 4			3238

temperature load influence factors, and aging influence factors are constructed as inputs to the model:

$$H^0 = [\delta_H, \delta_S, \delta_T]W^0 \tag{2}$$

Where δ_H, δ_S, δ_T denote the set of hydraulic load influencing factors , temperature influence factors, and aging influence factors, H^0 is the environment feature input to the graph convolutional feature extractor, and $W^0$is the trainable weight matrix.

4.2.2. Graph convolutional feature extractor

To better extract the features and inter-coupling futures between various hydraulic load influence factors, temperature load influence factors, and time-effect influence factors, the GCN [51] is used to construct the feature extractor. Specifically, the GCN represents the influence factor graph as $G = (V, E)$, where V denotes the set of influence factor nodes, and E denotes the set of edges. For each influence factor node, its feature representation is updated by a weighted sum of the features of its neighboring influence factor nodes. This weighted summation operation can be considered as aggregating the local neighborhood features. At each layer, the graph convolutional feature extractor updates node features using the following propagation rule [52]:

$$H^{(l+1)} = \sigma\left(\widetilde{D}^{-\frac{1}{2}}\widetilde{A}\widetilde{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}\right) \tag{3}$$

Where $H^{(l+1)}$ represents the feature matrix of the influence factor nodes at the l-th layer, $\widetilde{A} = A + I$ is the influence factor adjacency matrix with self-connections, $\widetilde{D}$ is the normalized degree matrix, $W^{(l)}$is the trainable weight matrix, and σ is the nonlinear activation function.

The seepage and deformation processes of high earth-rock dams are interrelated and represent the primary physical development processes of the dam [20]. To achieve a digital twin of the high earth-rock dam, it is essential to accurately map its seepage and deformation processes.

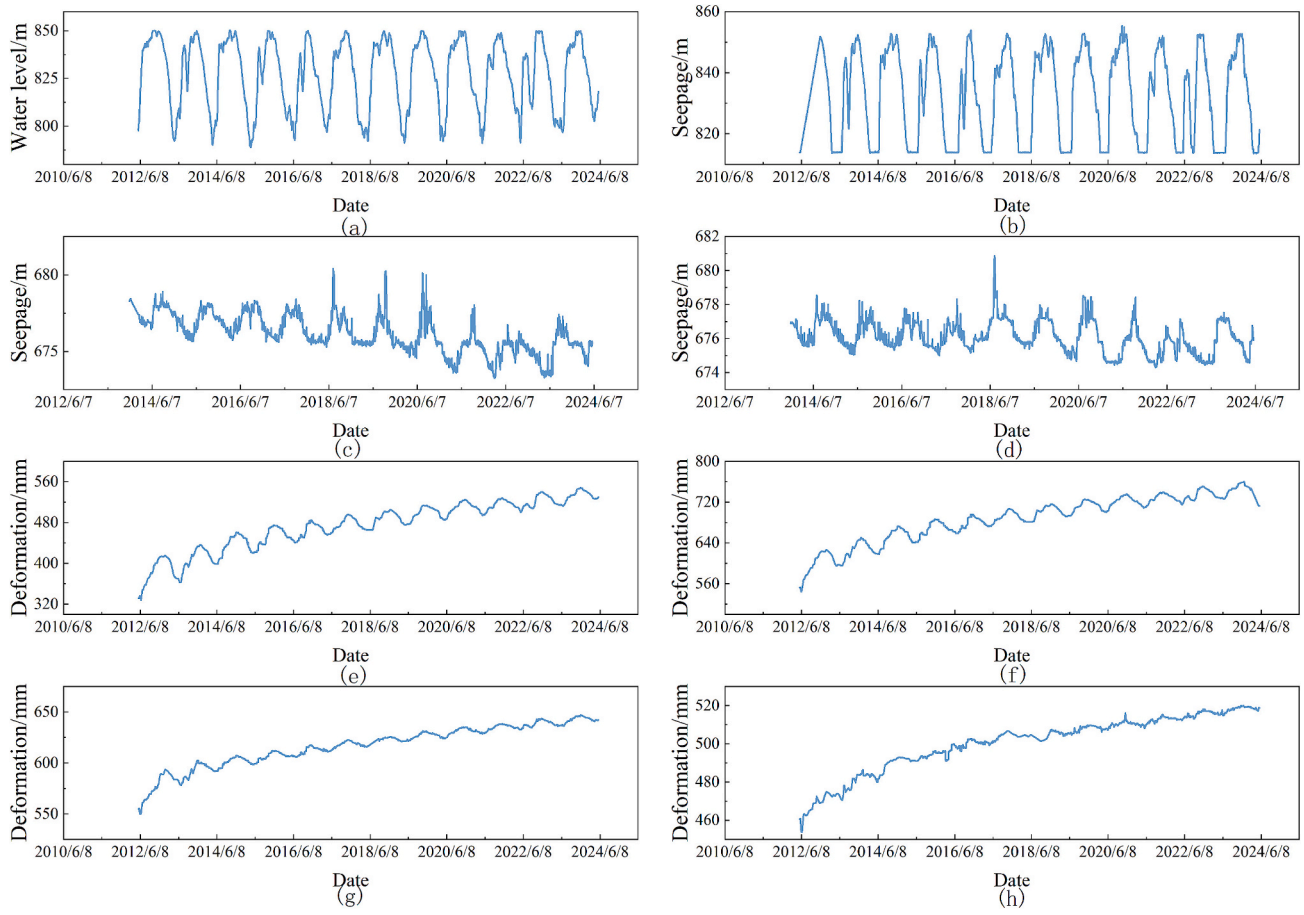


Fig. 5. Monitoring the process line of measured values at measurement points. (a) Water level (b) P10 (c) P29 (d) P32) (e) CH1 (f) CH4 (g) CH8 (h) CH16.

Therefore, integrating sensor data from seepage and deformation measurement points becomes critical. To enable real-time fusion of seepage and deformation monitoring features, this paper adopts a pre-training and feature fusion approach for integration. The graph convolutional feature extractor is used to separately construct pre-training extractors for seepage features and deformation features, extracting environment-seepage influence features $H_{Seepage}^{(l+1)}$ and environment-deformation influence features $H_{\text{Deformation}}^{(l+1)}$. The feature fusion is then realized by concatenating these features:

$$H_{\text{Coupling}}^{(l+1)} = \left[H_{\text{Seepage}}^{(l+1)}, H_{\text{Deformation}}^{(l+1)}\right] \tag{4}$$

4.2.3. Loss functions for deformation features extractor

Considering the poor extrapolation capability of deep learning models for extreme water level conditions with missing monitoring data, knowledge distillation technology is used. The hydraulic load teacher model and soil-rock creep teacher model trained with the FEM apply physical constraints to the deformation feature pre-training extractor during the training process. To simultaneously supervise the learning of the hydraulic load features and soil-rock creep features of the earth-rock dam, the deformation features extractor loss functions L_{MYD} in the pre-training process can be expressed as [26]:

$$L_{MYD} = \alpha L_{task}(l_S, t) + (1-\alpha)L_{KD}\left(p(f_s, T), E\left(p\left({f_T}^H, T\right), p\left({f_T}^\theta, T\right)\right)\right) \tag{5}$$

$$L_{KD} = DF\left(p\left(f_S, T\right), p(f_T, T)\right) \tag{6}$$

$$L = \alpha L_{task} + (1-\alpha)L_{KD} \tag{7}$$

$$p(f, T) = solfmax(f/T) \tag{8}$$

Where:L_{task} denotes the loss predicted by the student network regression of the pre-trained extractor of deformation features, L_{KD} denotes the gap between the teacher network and the student network, $DF(\cdot)$ denotes the distance metric function, f_S denotes the student network features, f_T denotes the teacher network features, T is the temperature parameter, and α is the loss balance parameter. l_S denotes the predicted value of deformation for the student network, t denotes the measured value of deformation, $E(\cdot)$ denotes the integrated output function to the teacher network, ${f^H}_T$ denotes the hydraulic loading features of the teacher network, and ${f^\theta}_T$ denotes the soil and rock creep features of the teacher network.

4.2.4. Time-varying effect update for decoder

The decoder is the component of a deep learning model responsible for generating prediction results from the feature extraction output [57]. It plays a crucial role in many sequence generation tasks. The seepage-deformation coupling feature $H_{\text{Coupling}}^{(l+1)}$ uses a shared, trainable feature decoder to simultaneously output the seepage monitoring prediction $\widehat{\delta}_{\text{Seepage}}$ and the deformation monitoring prediction $\widehat{\delta}_{\text{Deformation}}$:

$$\left[\widehat{\delta}_{\text{Seepage}}, \widehat{\delta}_{\text{Deformation}}\right] = \sigma\left(W_{decoder}H_{\text{Coupling}}^{(l+1)} + b_{decoder}\right) \tag{9}$$

Where $W_{decoder}$ is the trainable weight matrix, $b_{decoder}$ is the bias, and σ is the nonlinear activation function.

The time-varying effects of parameters caused by factors such as particle breakage in high earth-rock dams result in deformation monitoring curves exhibiting significant trends. Traditional machine learning models typically assume that training and testing data come from the

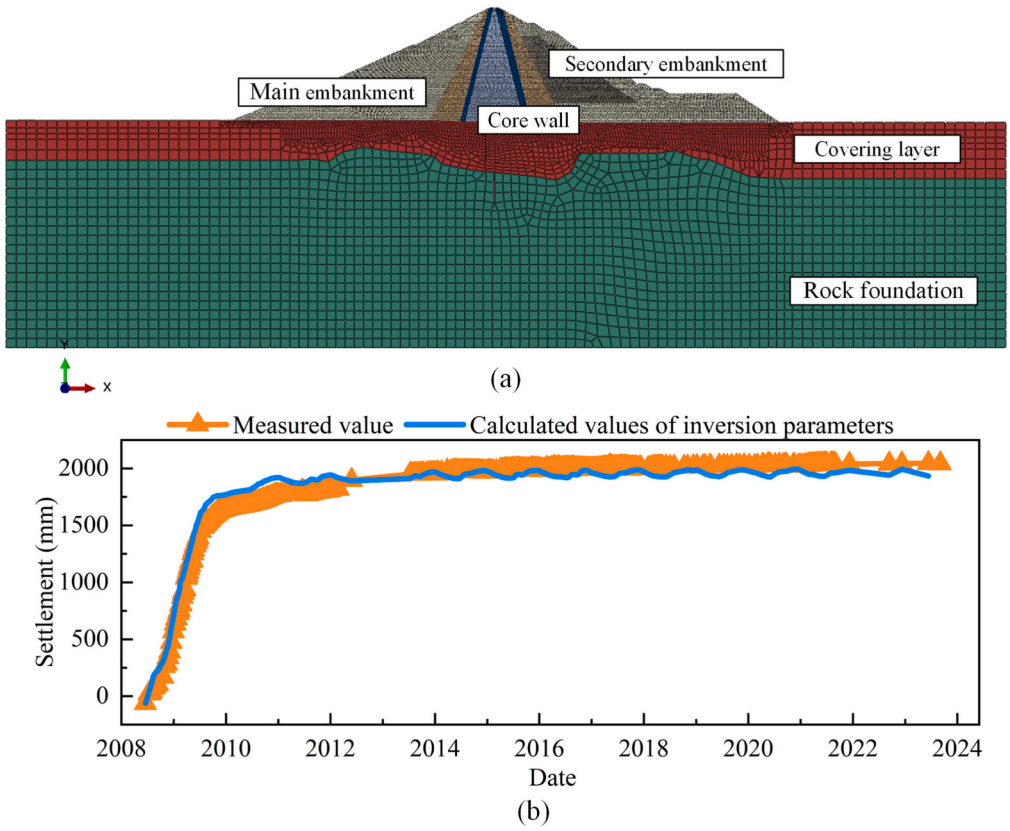


Fig. 6. Material zoning at section 0 + 210 of the dam and parameter inversion results. (a) FEM meshing and material partitioning. (b) Comparison of calculated and measured values of the settlement measurement point.

Table 2
Parameters of the seven-parameter creep model of the PB dam.

Parameters	α	b	c	d	m_1	m_2	m_3
Main embankment	9.86*E*-03	5.18E-05	5.18E-05	1.12E-03	5.14E-01	4.16E-01	4.27E-01
Secondary embankment	6.37E-03	5.72E-05	5.72E-05	2.47E-03	8.48E-01	4.55E-01	5.42E-01
Core wall	6.32E-03	1.00E-04	1.00E-04	3.01E-03	9.96E-01	6.79E-01	5.18E-01
Covering layer	9.86E-03	5.18E-05	5.18E-05	1.12E-03	8.48E-01	4.55E-01	5.42E-01

Table 3
Parameters of the Duncan-Zhang E-B model of the PB Dam.

Parameters	φ	*Rf*	*Ke*	*Kb*	*m*	*n*
Main embankment	44.4	0.7	1350	668.35	0.22	0.2
Secondary embankment	42.1	0.7	796	421.34	0.22	0.2
Core wall	35.0	0.8	384	207.70	0.52	0.4
Covering layer	38.0	0.7	1350	889.25	0.27	0.2

Table 4
Permeability coefficient of the PB dam.

Material zoning	Permeability coefficient *K* (cm/s)
Covering layer	7.50E-02
Upper zone of the core wall	6.94E-05
Down zone of the core wall	3.47E-05
Transition layer	3.00E-02
Main embankment	1.00E-01
Secondary embankment	1.00E-01
Rock foundation	1.00E-05
Filter layer	8.00E-03
Concrete seepage barrier wall	1.00E-07

Table 5
Main parameter settings of the models in the paper.

Model	Main Parameter Setting
HST	/
FEM	/
SVM	penalty parameter = 5, epsilon = 0.01
LSTM	hidden layer dimension = 64
GRU	hidden layer dimension = 64
GCN	hidden layer dimension = 64
Transformer	hidden layer dimension = 64, encoder layer = 2, decoder layer = 2, attention head = 16
TGCN	hidden layer dimension = 64
Ablation model	hidden layer dimension = 192
Cell body model	hidden layer dimension = 64
Synapse model	hidden layer dimension = 192

same independent and identically distributed space. However, over time, this assumption may lead to a significant decline in the predictive performance of monitoring models for high earth-rock dams. To address this, a parameter-sharing fine-tuning transfer learning technique is applied to fine-tune the feature decoder, improving the model's generalization ability [2,53]:

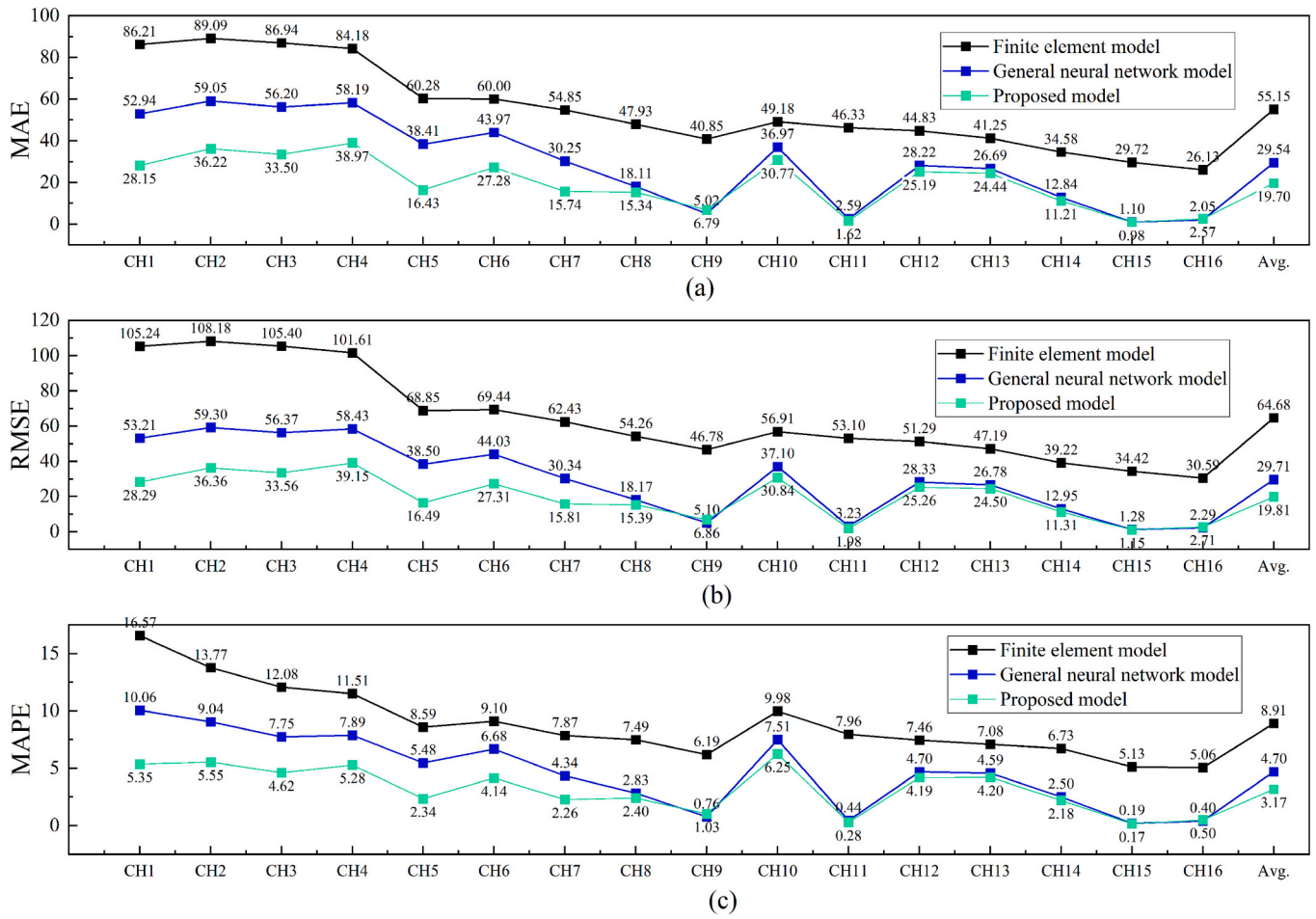


Fig. 7. Comparison of the accuracy of deformation results produced by the FEM, the ablation model, and the proposed model with the actual monitored values. (a) MAE (b) RMSE (c) MAPE.

$$\theta_{shared}(W_{decoder}, b_{decoder}) = \theta_{pretrain}(W_{decoder}, b_{decoder}) \tag{10}$$

$$\theta_{update} = \underset{\theta}{argmin} L_{Recent\ monitoring\ data}\left(f_{Cell\ body}([\delta_H, \delta_S, \delta_T]; \theta_{shared})\right) \tag{11}$$

Where $\theta_{pretrain}$ is the trained parameters of the feature decoder, θ_{shared} and θ_{update} are the shared and updated parameters , $f_{Cell\ body}(\cdot)$ is the pre-training cell body model, and $L_{Recent\ monitoring\ data}$ is the loss function for the task of updating the recent monitoring data.

4.3. Synapse model

The synapse model principle is illustrated in Fig. 3. This section presents the synapse model, which employs an LSTM network to learn information transmission relationships among measurement points. A loss function is designed to optimize the transformation from sensor data to a digital twin, with a dynamic monitoring loss to adjust for FEM discrepancies. Finally, the model integrates seepage and deformation predictions to construct a comprehensive digital twin model for high earth-rock dams.

4.3.1. Network structure

Considering a large number of measurement points, an LSTM is employed to learn the node information transmission relationship. LSTM [54] is a type of RNN specifically designed to handle sequence data, addressing the vanishing and exploding gradient problems that occur with traditional RNNs when learning long sequences. The key feature of LSTM lies in its internal structure, which includes three gates: the input gate, the forget gate, and the output gate. These gates control the writing, retention, and output of information, respectively. The core computational formulas of the network structure for the synapse model are as follows [55]:

$$f_t = \sigma\left(W_f \cdot \left[h_{t-1}, \delta_t^{Point}\right] + b_f\right) \tag{12}$$

$$i_t = \sigma\left(W_i \cdot \left[h_{t-1}, \delta_t^{Point}\right] + b_i\right) \tag{13}$$

$$o_t = \sigma\left(W_o \cdot \left[h_{t-1}, \delta_t^{Point}\right] + b_o\right) \tag{14}$$

$$C_t = f_t^{*} C_{t-1} + i_t^{*} \widetilde{C}_t \tag{15}$$

$$h_t = o_t^{*} tanh(C_t) \tag{16}$$

$$\delta_t^{ALL} = \sigma(W \cdot h_t + b) \tag{17}$$

Where f_t, i_t, and o_t are the activation values of the forgetting, input, and output gates, respectively, C_t is the state of the memory cell at the current moment, h_t is the output state, δ_t^{Point} is the input point-of-measurement value, W and b are the learnable weight and bias, respectively, and δ_t^{All} is the predicted value of the all digital twin node.

4.3.2. Loss function

To realize the transformation of the measurement point monitoring model to the digital twin model, the synapse model is used to learn the seepage pressure or deformation values of finite element nodes δ_{Point}^{FEM} corresponding to the sensor monitoring points, and the information transfer relationship between the sensors' actual monitoring values

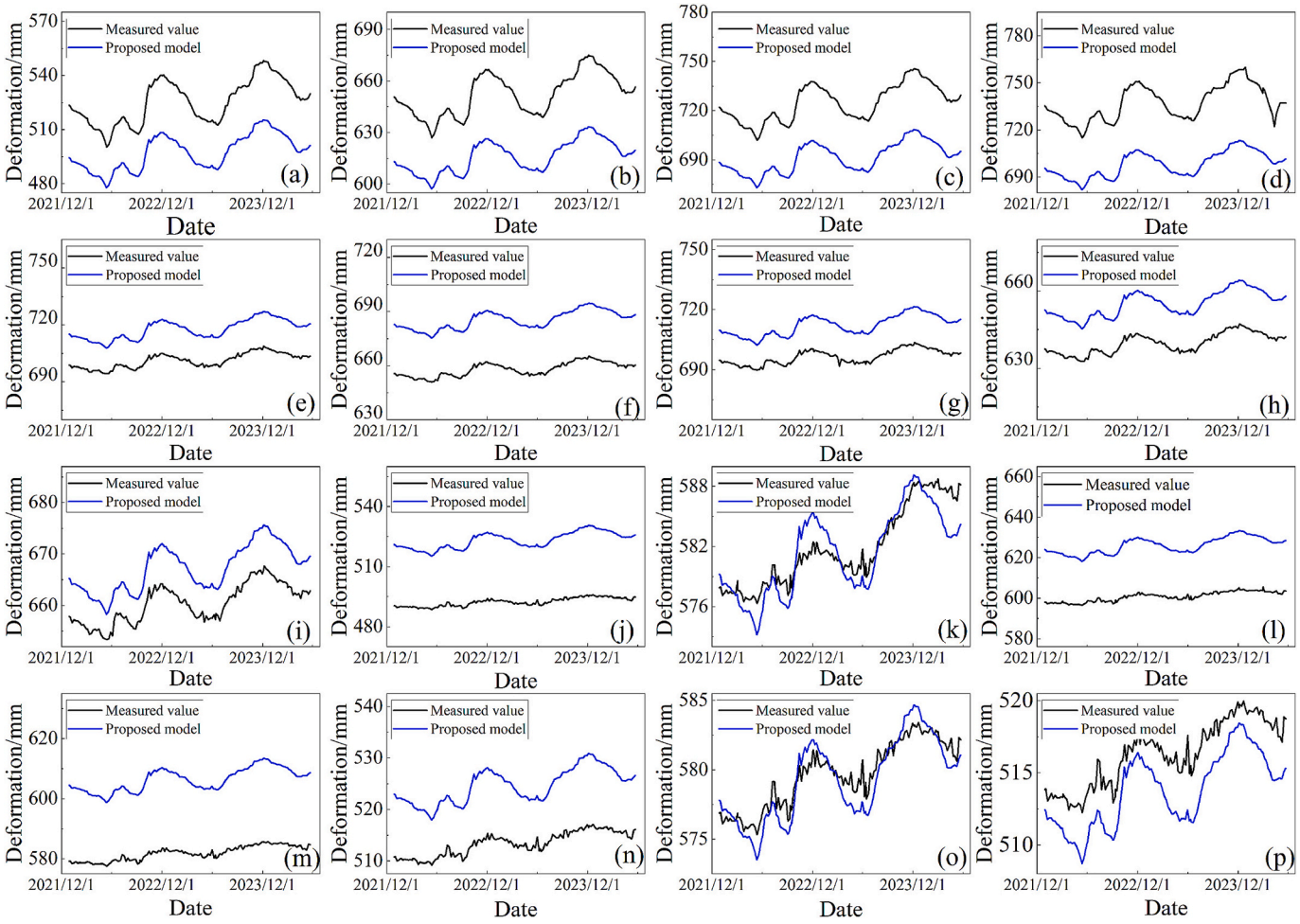


Fig. 8. Comparison of predicted value process lines with measured value process lines of the proposed model on the test set. (a) CH1 (b) CH2 (c) CH3 (d) CH4 (e) CH5 (f) CH6 (g) CH7 (h) CH8 (i) CH 9 (j) CH10 (k) CH11 (l) CH12 (m) CH13 (n) CH14 (o) CH15 (p) CH16.

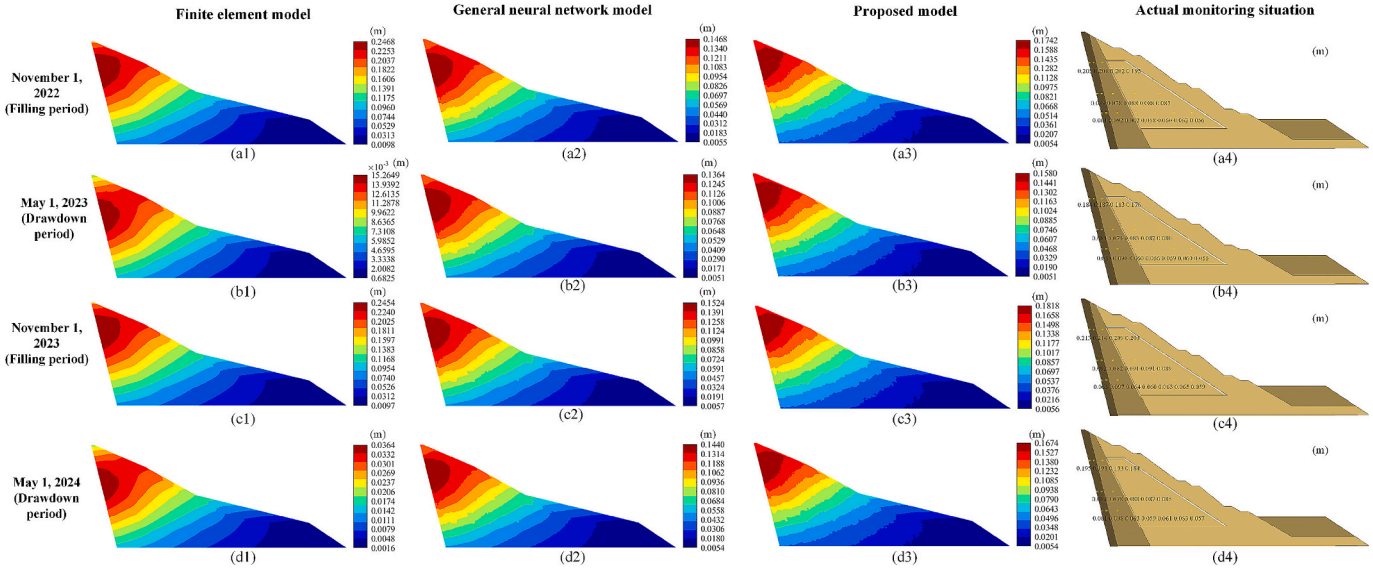


Fig. 9. Comparison of cross-section deformation cloud maps predicted by the FEM, the ablation model, and the proposed model, as well as the actual monitoring at four typical moments. (a) November 1, 2022 (b) May 1, 2023 (c) November 1, 2023 (d) May 1, 2024.

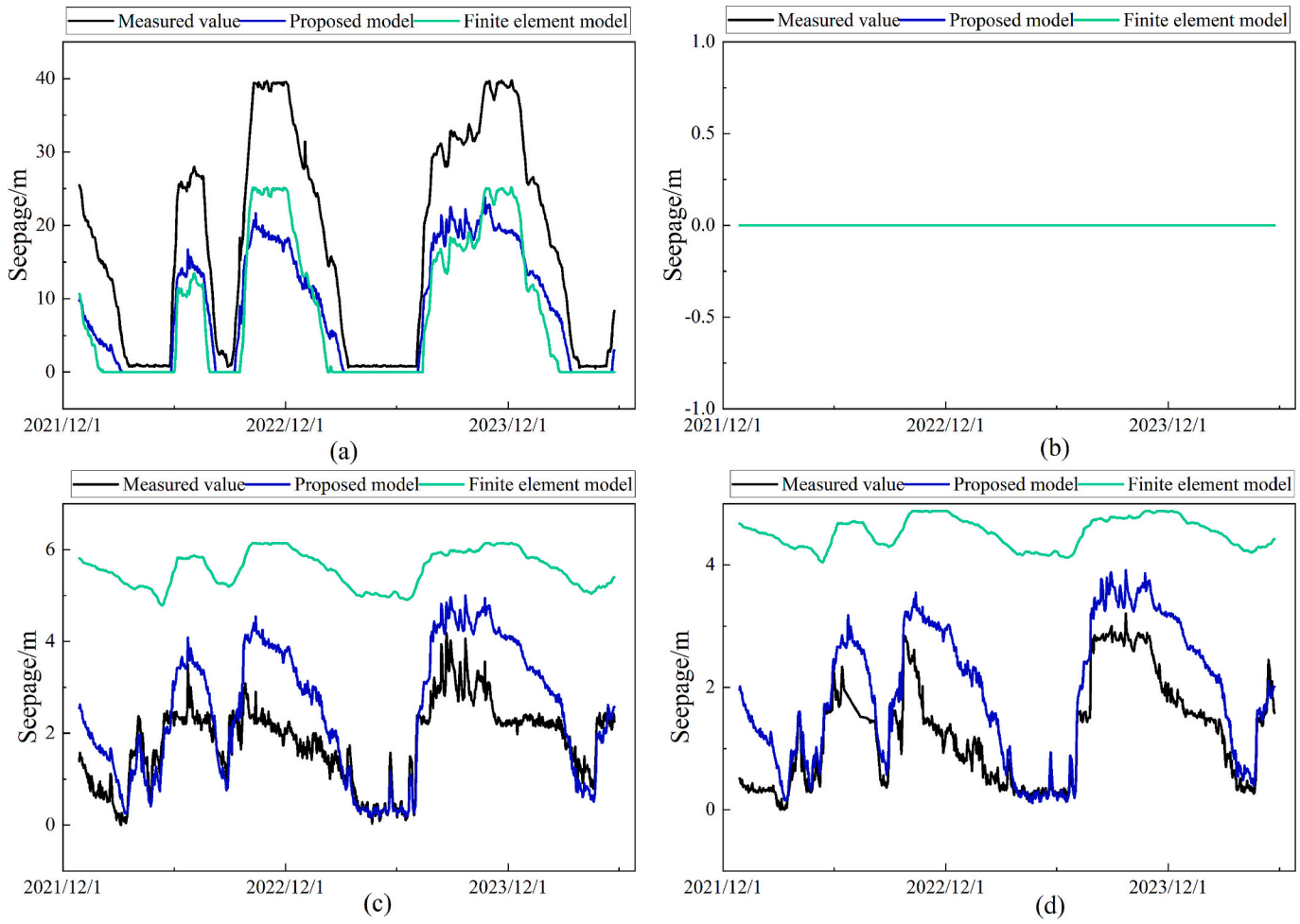


Fig. 10. Comparison of predicted versus measured value process lines for the FEM and the proposed model for all seepage pressure measurement points on the test set. (a) P10 (b) P22 (c) P29 (d) P32.

$\delta_{Point}^{Monitoring}$ and the seepage pressure or deformation values of all finite element nodes δ_{All}^{FEM} to obtain the pre-trained mapping model:

$$\theta_{Digital\ twin} = argmin L_\theta\ Digital\ twin\left[L_{ALL}^{FEM}\left(f_{\mathbf{Synapse}}\left(\delta_{Point}^{FEM};\theta\right)\right), L_{Point}^{FEM}\left(f_{\mathbf{Synapse}}\left(\delta_{Point}^{FEM};\theta\right)\right), L_{Point}^{Monitoring}\left(f_{\mathbf{Synapse}}\left(\delta_{Point}^{Monitoring};\theta\right)\right)\right] \tag{18}$$

Where $\theta_{Digital\ twin}$ is the pre-trained parameters of the synapse model, θ is the parameters of the synapse model, $f_{Synapse}(\cdot)$ is the synapse model, $L_{Digital\ twin}$ is the loss of the synapse model, L_{Point}^{FEM} is the loss of the corresponding nodes to δ_{Point}^{FEM}, $I_{Monitoring}^{Point}$ is the loss of the nodes corresponding to the actual monitoring points $\delta_{Point}^{Monitoring}$, and L_{All}^{FEM} is the loss of all finite element nodes.

Considering the difference between the FEM and the actual monitoring situation, a dynamic monitoring loss function is proposed to adjust the learning of the synapse model to the actual association situation of the nodes during the model training process:

$$L_{Digital\ twin} = \alpha\left(L_{Point}^{FEM} + L_{Point}^{Monitoring}\right) + (1-\alpha)L_{All}^{FEM} \tag{19}$$

$$L_{Point}^{FEM} = \frac{1}{N}\sum_{i=1}^{N} l\left(\delta_{Point}^{FEM}, f_{Synapse}\left(\delta_{Point}^{FEM};\theta\right)\right) \tag{20}$$

$$L_{Point}^{Monitoring} = \frac{1}{N}\sum_{i=1}^{N} l\left(\delta_{Point}^{Monitoring}, f_{Synapse}\left(\delta_{Point}^{Monitoring};\theta\right)\right) \tag{21}$$

$$L_{All}^{FEM} = \frac{1}{N}\sum_{i=1}^{N} l\left(\delta_{All}^{FEM}, f_{Synapse}\left(\delta_{Point}^{FEM};\theta\right)\right) \tag{22}$$

Where $l(\cdot)$ is the error term (e.g., mean square error or cross-entropy), α is the loss balance parameter.

4.3.3. Model connection

The digital twin model can be obtained by inputting the predicted values of the seepage process and deformation process of high earth-rock dams by the cell body model into the synapse model:

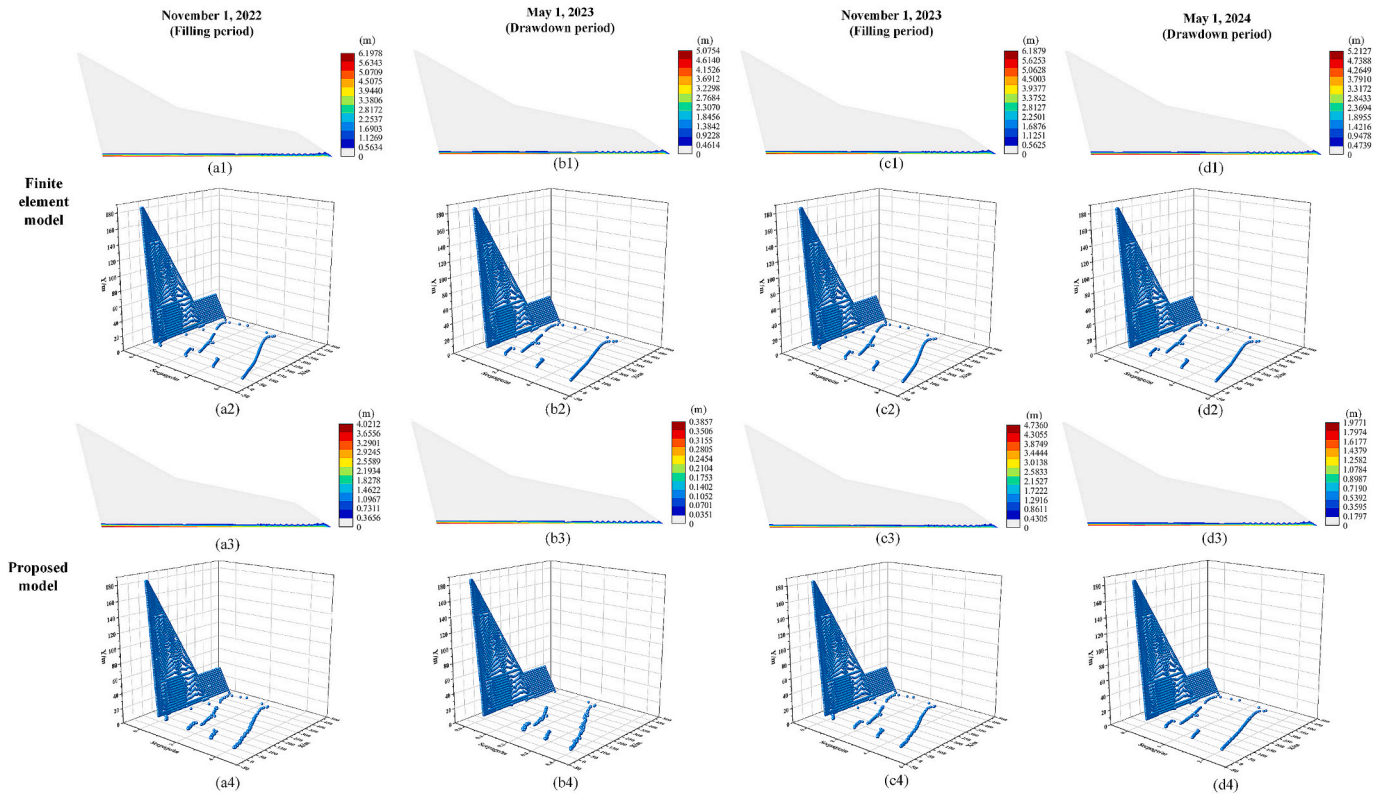


Fig. 11. Comparison between the FEM and the proposed model on the test set for four typical moments of predicted seepage pressure clouds in the section. (a) November 1, 2022 (b) May 1, 2023 (c) November 1, 2023 (d) May 1, 2024.

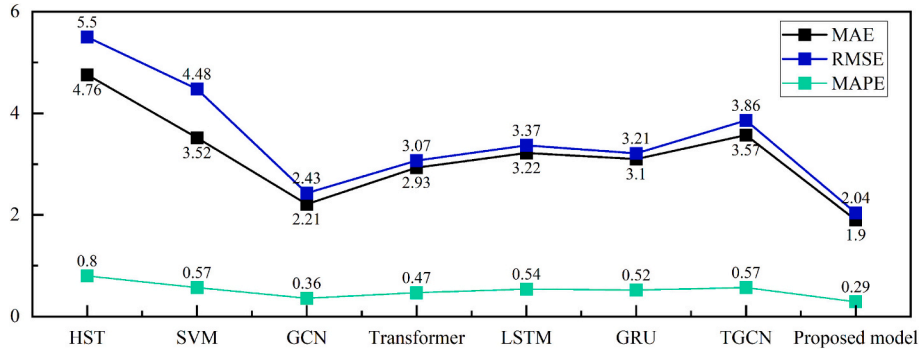


Fig. 12. Comparison of the average MAE(mm), RMSE(mm), and MAPE(%) between the cell body model and the baseline model for all measurement points.

$$\left[\widehat{\delta}_{Seepage}, \widehat{\delta}_{Deformation}\right] = f_{Cell\ body}\left(\left[\delta_H, \delta_S, \delta_T\right]; \theta_{Update}\right) \tag{23}$$

$$\delta_{Digital\ twin} = f_{Synapse}\left(\left[\widehat{\delta}_{Seepage}, \widehat{\delta}_{Deformation}\right]; \theta_{Digital\ twin}\right) \tag{24}$$

5. Project case

This section analyzes the PB Hydroelectric Station using monitoring data and finite element modeling. It examines displacement and seepage trends, develops an FEM for structural analysis, and evaluates model accuracy with key metrics. Various baseline models are compared to validate the proposed model.

5.1. Project overview

The PB Hydroelectric Station is a large-scale hydropower project primarily focused on power generation, with additional benefits in flood control, sediment retention, and other integrated uses. The dam type is a gravel-filled earth-core rockfill dam, with a top elevation of 856 m, a maximum dam height of 186 m, a normal reservoir water level of 850 m, and a dead water level of 790 m. As shown in Fig. 4, this paper focuses on the typical section of PB Dam at the elevation 0 + 210, where a group of horizontal displacement meters using tensile lines is arranged at the downstream filter layers, transition layers, and rockfill body at elevations 808 m, 758 m, and 731 m. These meters are numbered from CH1 to CH16, with a total of 16 measurement points. Currently, there are three monitoring points for seepage pressure on the downstream dam slope of this section, numbered P22, P29, and P32. The remaining measurement points are either out of service or have been discontinued.

5.2. Data sets

This paper creates a dataset using deformation and seepage monitoring data from the PB Dam's 0 + 210 section, spanning from May 21, 2012, to May 23, 2024, along with the finite element simulation data.

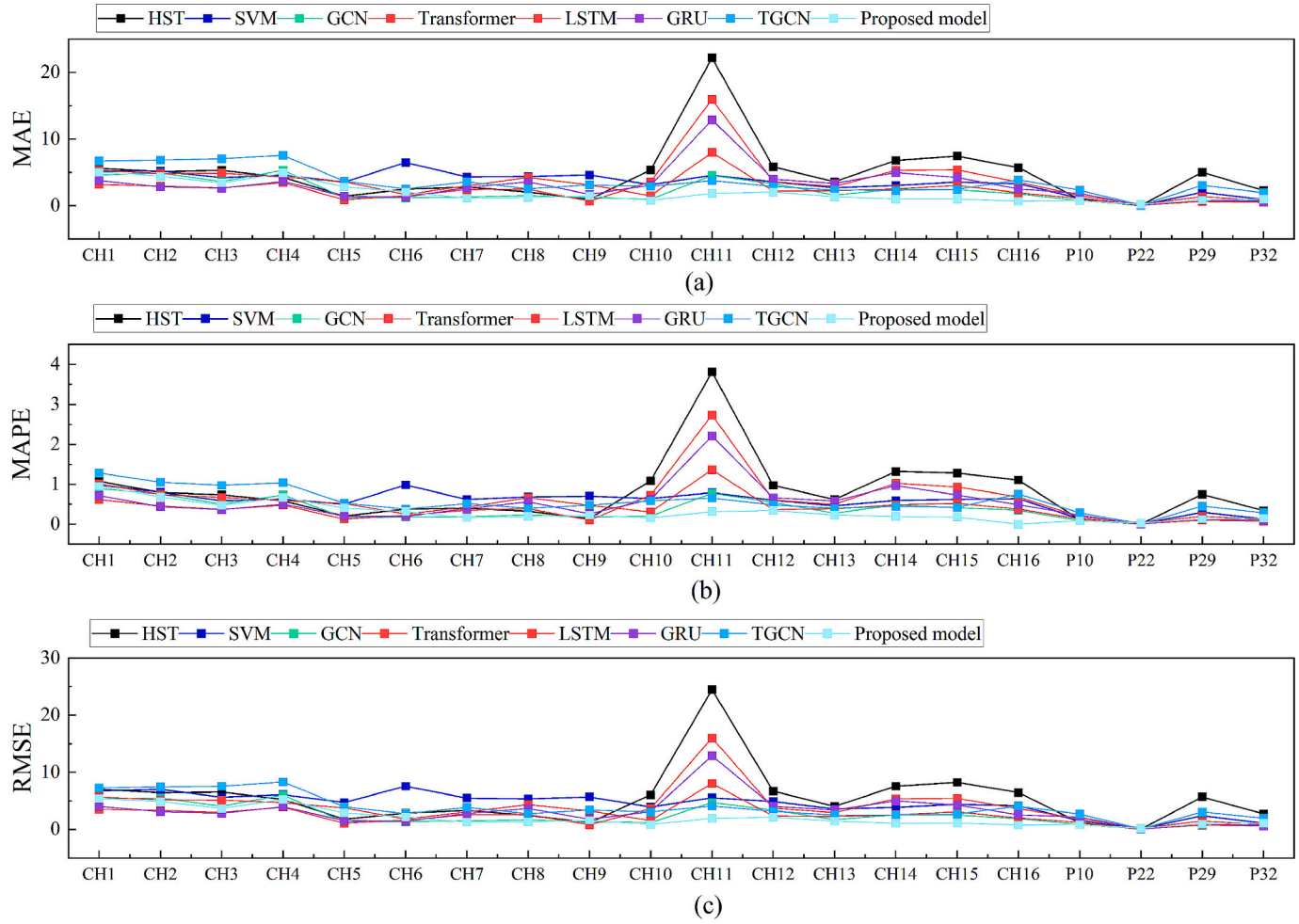


Fig. 13. Comparison of the specifics of MAE, RMSE, and MAPE between the cell body model and the baseline model for all measurement points. (a) MAE(mm) (b) MAPE(%) (c) RMSE(mm).

The training and testing data division ratios are shown in Table 1. The hydraulic influence factors (H, H^2, H^3, H^4), temperature influence factors$(cos(\varpi t), sin(\varpi t), cos(\varpi t)*sin(\varpi t))$, and aging influence factors$(\theta, ln(\theta), e^{-\theta})$ of all model inputs are consistent [56], where the calculations of the impact factors have been described previously. The process lines of the measured reservoir water levels and the values of some deformation and seepage monitoring points are shown in Fig. 5. It can be observed that, due to the annual regulation of the reservoir water level, all measurement points exhibit periodic variations influenced by the water level. The filling period typically occurs between early and late October, but in dry years, it can be delayed until mid-November or later. The drawdown period is primarily concentrated in mid-to-late April. Additionally, it can be seen that the seepage pressure measurement points exhibit changes that are largely consistent with the reservoir water level variations, while the deformation measurement points show a distinct trend due to the effects of creep.

5.3. Finite element model

In the FEM, the actual conditions such as the dam's partitioning layout and the thick covering layer of the foundation are taken into account. Additionally, based on the results of in-situ drilling tests, the model divides the dam into an upper and lower zone, with the dead water level at 790 m as the boundary. The simulation range extends 2 times the dam height upstream, downstream, and below the dam foundation. The model consists of 6572 elements and 13,348 nodes, with the finite element mesh and material division shown in Fig. 6(a). The PB dam is divided into the upstream embankment zone, transition layer, filter layer, gravel soil core wall, downstream secondary embankment zone, and downstream main embankment zone. This paper focuses on the transient and creep key parameter inversion of the main embankment, secondary embankment, core wall, and covering layer of the typical section. The parameter values for these are shown in Table 2 and Table 3, while the designed permeability coefficients for each partition of the PB dam are shown in Table 4. From Fig. 6(b), it can be observed that the settlement curve calculated by the FEM matches well with the measured values of the monitoring points.

5.4. Evaluation methods

To quantitatively analyze the evaluation performance of each model, three evaluation metrics Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE) are chosen in this paper to evaluate the difference between real and predicted flows. The formula for the evaluation indicators below, $\widehat{Y}_t$ denotes the predicted value, Y_t denotes the true value, $\overline{Y}$ denotes the average of the true values, and n is the number of true values.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|Y_t - \widehat{Y}_t| \tag{25}$$

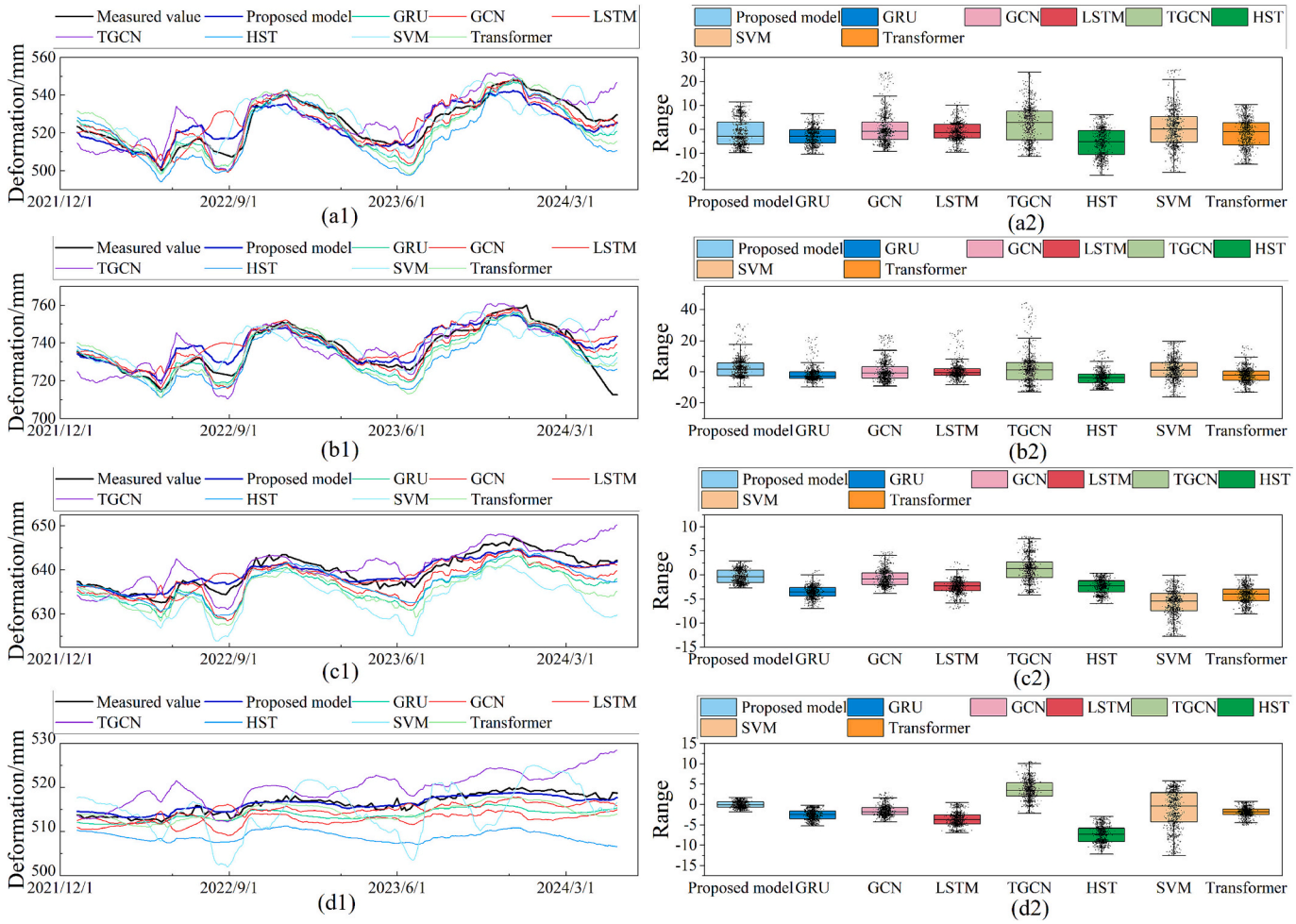


Fig. 14. Comparison of predicted and measured value process lines for the cell body model and the baseline model for some of the deformation measurement points on the test set. (a) CH1 (b) CH4 (c) CH8 (d) CH16.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(Y_t - \widehat{Y}_t)^2} \tag{26}$$

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\frac{(Y_t - \widehat{Y}_t)}{Y_t} \times 100\% \tag{27}$$

5.5. Model parameters

To more accurately assess the performance of the proposed model, we selected the following nine baseline models for comparison: the classical statistical model HST [41], the FEM, five baseline deep learning models (LSTM [16], GRU [18], GCN [19], Transformer [57], TGCN [58]), a shallow machine learning model (SVM [12]), and an ablation model. The ablation model uses only the LSTM for the synapse model. The key parameter settings for these models are shown in Table 5.

6. Analysis of results

This experiment is conducted under the Windows 10 system environment using Pycharm IDE and Pytorch2.5.1 based on Python3.10. The server configuration follows: CPU is Intel Core i7–13,700, GPU is Nvidia RTX 4070SUPER. In the experiment, the learning rate is set to 0.001, the batch size is 64, the optimizer selects the Adam optimizer to update the network parameters, and the number of iterations is set to 500.

6.1. Analysis of the accuracy of the proposed model

To validate the accuracy of the deformation and seepage pressure results generated by the proposed model for the cross-section of high earth-rock dams, the predicted values at sensor-monitored nodes were compared with the actual measured values, as well as with the FEM results and predictions from the ablation model of the proposed model. It is important to note that, to simplify numerical simulation, the deformation results compared here represent incremental displacements relative to the baseline measurement taken on May 21, 2012.

As shown in Fig. 7, the MAE, RMSE, and MAPE of predictions for all deformation monitoring points in the 0 + 210 cross-section were compared among the three models. The proposed model achieved the best accuracy, with average improvements over the FEM of 64.28 %, 59.37 %, and 64.42 % in terms of MAE, RMSE, and MAPE, respectively. This demonstrates that integrating actual monitoring data significantly enhances the simulation accuracy of the FEM. Similarly, compared to the ablation model using only the LSTM, the proposed model improved the average accuracy of all monitoring points by 33.31 %, 33.32 %, and 32.55 % in terms of MAE, RMSE, and MAPE, respectively. This indicates that the dynamic monitoring loss function effectively integrates and mutually corrects the FEM with actual monitoring data. From the perspective of individual monitoring points, the proposed model consistently achieved the best prediction accuracy across all points, highlighting its strong generalization capability. Furthermore, the performance improvement was particularly pronounced for monitoring points near the dam crest compared to the other two models. This

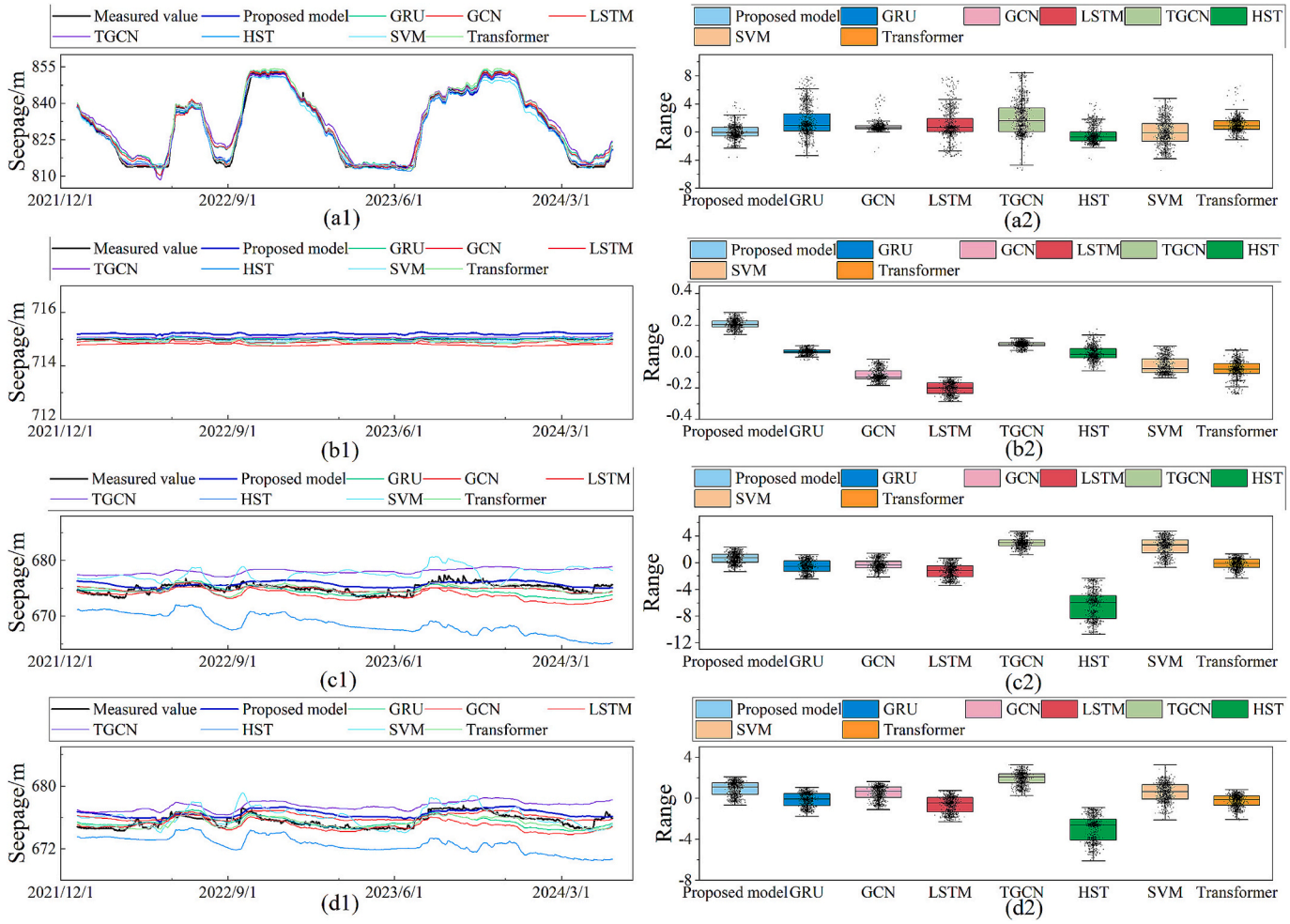


Fig. 15. Comparison of predicted and measured value process lines between the cell body model and the baseline model for seepage pressure measurement points on the test set. (a) P10 (b) P22 (c) P29 (d) P32.

suggests that deformation in the upper part of the dam is more significantly influenced by factors such as water levels, making it more challenging to simulate and predict accurately, which aligns well with practical engineering observations.

Fig. 8 illustrates the comparison between the predicted value trends of the proposed model and the measured value trends for all monitoring points CH1-CH16 in the test set. It should be noted that the prediction results of the FEM and the ablation model are not shown due to their significant deviations from the measured values. From the figure, it can be observed that the trend of the proposed model's prediction curves aligns with the actual measurements. However, except for a few points, such as CH11 and CH15, which fit the measured values perfectly, most prediction curves are shifted by a certain distance relative to the measured curves. This offset represents the primary source of error in the proposed model, mainly attributed to discrepancies between the FEM's simulated initial displacement of the dam cross-section and the actual monitoring data. The average MAPE across all monitoring points for the proposed model is 3.17 %, demonstrating its applicability when compared to numerical simulation results. The largest prediction error occurred at CH10, with a deviation of 30.77 mm, while the smallest error was observed at CH15, at just 1.15 mm.

As shown in Fig. 9, the cross-section deformation contour plots predicted by the FEM, the ablation model, and the proposed model, along with the actual monitoring data, are compared at four typical times—November 1, 2022; May 1, 2023; November 1, 2023; and May 1, 2024—representing wet or dry seasons. The results show that the trends of horizontal deformation changes in the contour plots predicted by all three models are generally consistent, aligning with typical numerical simulation behaviors. This indicates that the FEM primarily determines the overall trend of cross-section deformation predicted by the proposed model. However, significant differences in the magnitude of deformation values are evident among the three models. The FEM's calculated horizontal displacements on November 1, 2022, and May 1, 2023, exhibit excessive changes, which deviate from actual conditions. In contrast, the predictions of the ablation model and the proposed model are closer to the actual monitoring data, demonstrating that incorporating real-world monitoring values can effectively correct the limited accuracy of the FEM's simulation. Moreover, the proposed model outperforms the other two models in terms of proximity to the actual monitoring data. This highlights its ability to effectively integrate the FEM with actual monitoring points, mutually compensating for each other's shortcomings.

Fig. 10 presents a comparison of the predicted time series for all piezometric points between the FEM and the proposed model against the actual measurements in the test set. It can be observed that at points P29 and P32, the proposed model effectively corrected the deviations from the actual monitoring data seen in the FEM, indicating that the proposed model demonstrates equally robust performance in piezometric predictions. Fig. 11 shows a comparison of the cross-section piezometric contour plots predicted by the FEM and the proposed model at four typical time points in the test set. The results reveal that the distribution patterns of piezometric values between the two models are generally

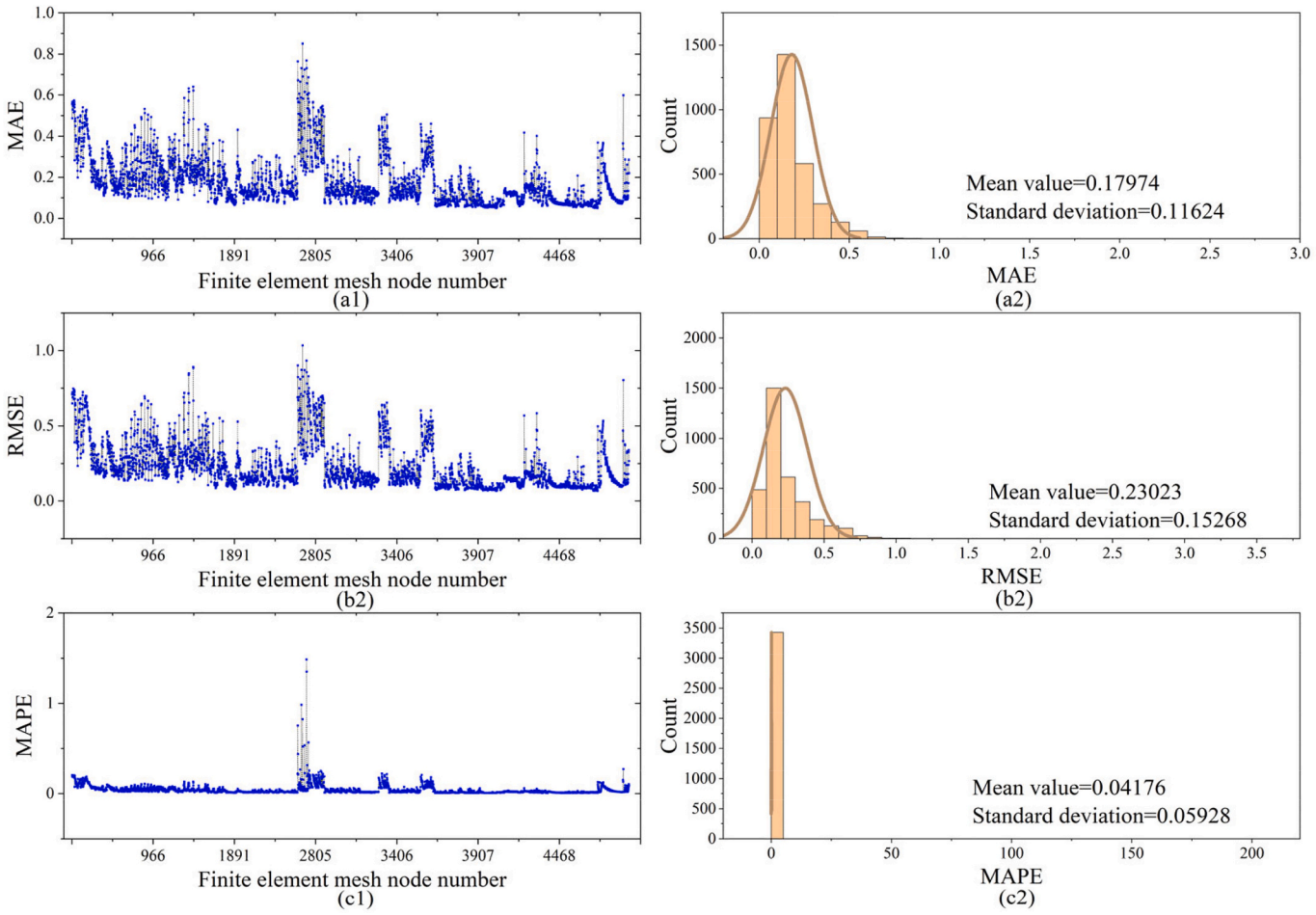


Fig. 16. Analytical results of MAE, RMSE, and MAPE for the synapse model for all nodes for the calculated values of horizontal displacements of the FEM. (a) MAE (mm) (b) RMSE(mm) (c) MAPE(%).

consistent, with only slight differences in the magnitudes of the values. This suggests that the proposed model effectively integrates the FEM with actual monitoring data, enabling real-time and accurate prediction of the two-dimensional piezometric conditions in high earth-rock dams.

6.2. Analysis of the accuracy of the cell body model

To analyze the primary sources of prediction errors in the proposed model and assess the effectiveness of the cell body model, the prediction accuracy of the cell body model for points CH1-CH16 and P10-P32 was compared with baseline models in terms of MAE, RMSE, and MAPE, as shown in Fig. 12 and Fig. 13. The results indicate that the proposed model achieved the best average prediction accuracy across all points, with improvements of 14.03 %, 16.05 %, and 19.44 % in MAE, RMSE, and MAPE, respectively, compared to the GCN model. This demonstrates that the proposed model which integrates seepage and deformation features, significantly enhances the prediction accuracy of the cell body model. Furthermore, compared to the results in Fig. 7, the MAE, RMSE, and MAPE of the cell body model decreased dramatically by 90.35 %, 89.70 %, and 90.85 %, respectively, suggesting that the prediction errors of the proposed model are minimally related to the cell body model itself. Fig. 13 further shows that the proposed model achieves the best prediction accuracy for the majority of monitoring points, highlighting its strong generalization capability.

Fig. 14 and Fig. 15 illustrate the comparison between the predicted value trajectories and the measured value trajectories of the cell model and baseline models for selected deformation and seepage pressure monitoring points on the test set, accompanied by corresponding residual analyses. The results show that the proposed model effectively captures the changing trends of both deformation and seepage pressure measurements, achieving high prediction accuracy. Compared with other models, the proposed model demonstrates superior predictive performance for the selected deformation and seepage pressure points. Moreover, the residual distribution is relatively stable, indicating that the proposed model exhibits robust predictive capabilities.

6.3. Analysis of the synapse model accuracy

To investigate the primary sources of error in the proposed model and evaluate the accuracy of the synapse model, the synapse model was used to fit the horizontal displacement, settlement displacement, and seepage pressure calculated by the FEM for 3238 nodes. The error results on the test set are presented in Fig. 16, Fig. 17, and Fig. 18. The analysis reveals that the synapse model achieves average MAPE values of only 0.041 %, 0.008 %, and 1.031 % for horizontal displacement, settlement displacement, and seepage pressure, respectively, indicating its capability to effectively fit the FEM's simulation of the cross-section of high earth-rock dams. Combined with the previous analyses, it can be concluded that the primary source of error in the proposed model stems from discrepancies between the trend variations in the FEM's simulations and the actual monitoring data. While the proposed model can partially correct the FEM's deviations by incorporating real monitoring data, its digital twin accuracy for the cross-section of high earth-rock dams remains fundamentally dependent on the FEM's simulation accuracy.

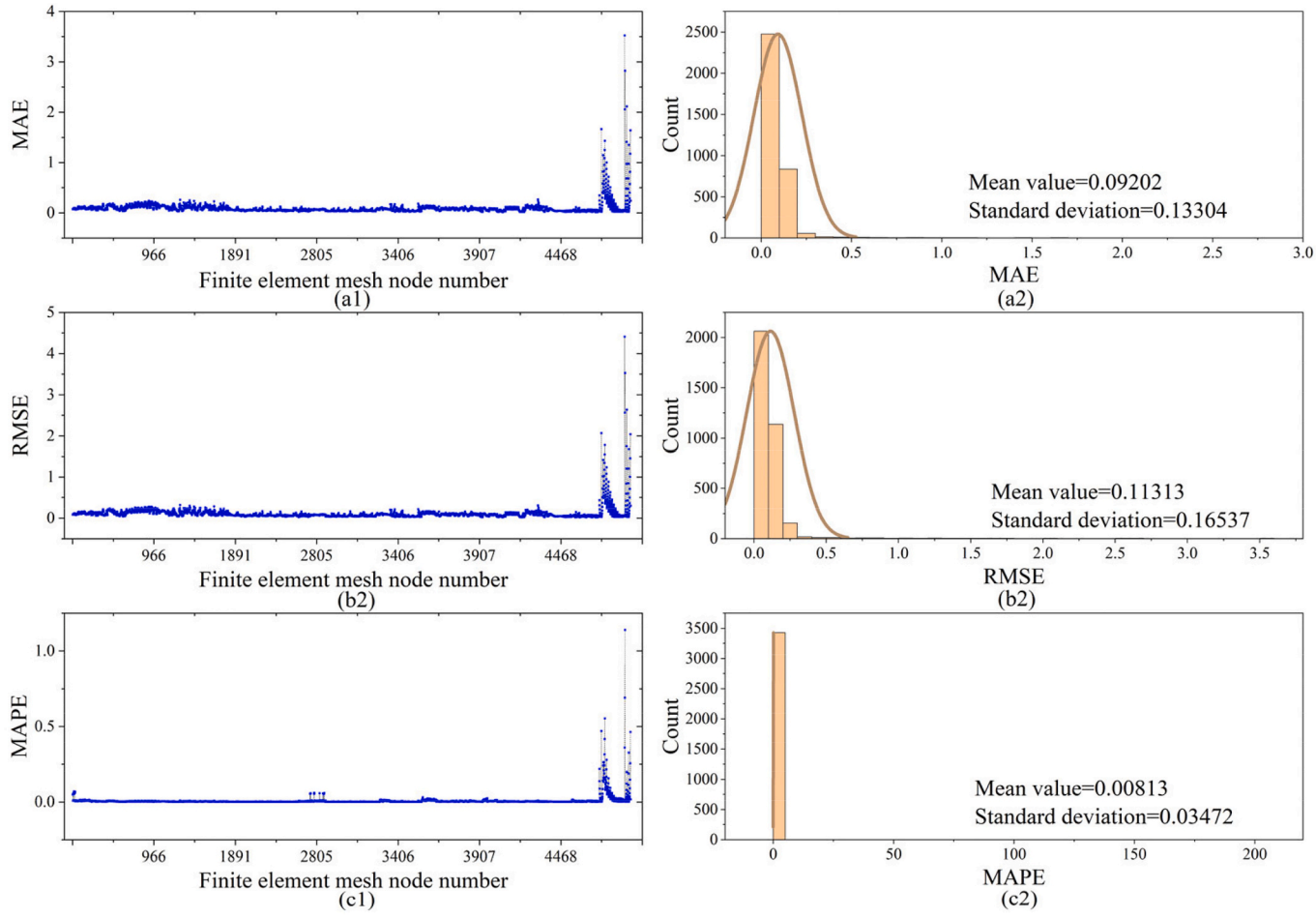


Fig. 17. Analytical results of MAE, RMSE, and MAPE for the synapse model for all nodes for the computed values of settlement displacements of the FEM. (a) MAE (mm) (b) RMSE(mm) (c) MAPE(%).

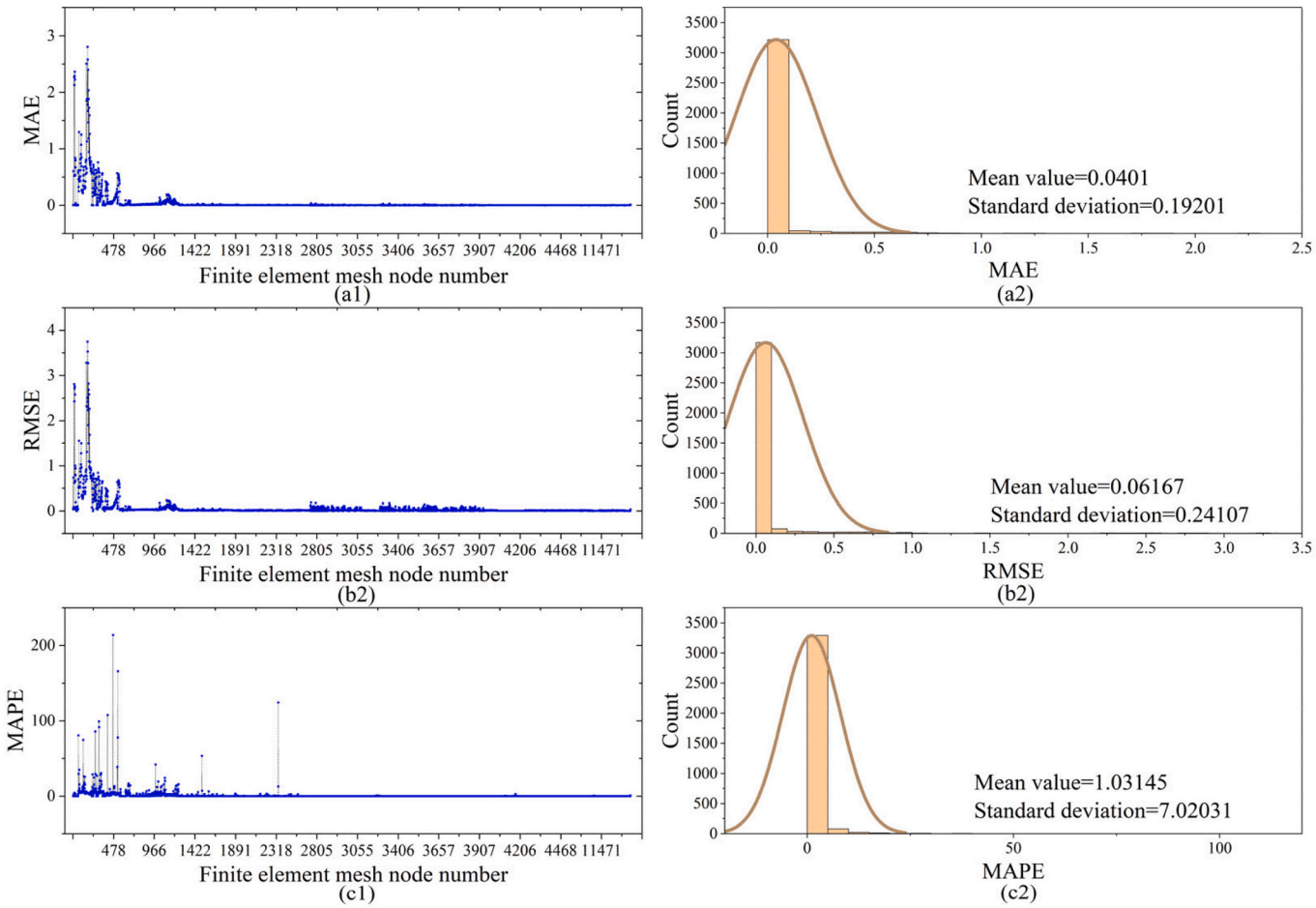


Fig. 18. Analytical results of MAE, RMSE, and MAPE for the synapse model for all nodes for the calculated values of percolation pressure of the FEM. (a) MAE(m) (b) RMSE(m) (c) MAPE(%).

7. Conclusions

This paper presented a deformation-seepage digital twin model for high earth-rock dams based on deep learning and FEM and applied it to a high earth-rock dam project. The main conclusions were as follows:

1. A monitoring point model was developed using GCN and transfer learning, enabling time-updating and extrapolation under extreme water level conditions. Furthermore, an innovative point-to-domain model integrated LSTM with the FEM to map monitoring point data onto two-dimensional cross-sectional information, with a dynamic loss function proposed to effectively fuse sensor data and FEM outputs.
2. The proposed model can effectively achieve the digital twin of the deformation-seepage for the cross-section of high earth-rock dams, with an average error of only 3.17 % between the predicted deformation and actual monitoring data. Compared with the traditional numerical simulation model, it can significantly improve simulation accuracy, with the deformation simulation accuracy increasing by 64.42 %. Furthermore, the dynamic monitoring loss method proposed in this paper can effectively improve the fusion prediction accuracy of the FEM and real monitoring data, with deformation prediction accuracy improving by 32.55 %.
3. The proposed prediction model for monitoring points achieved the best prediction accuracy and better generalization compared with baseline models. It was shown that the proposed deformation-seepage coupling prediction method can effectively improve the prediction accuracy of the synapse model by 19.44 %, as compared to the ablation model. Through accuracy analysis of the cell body model and synapse model, it was found that the main source of error in the proposed model arose from inconsistencies between the FEM simulation results and actual monitoring patterns. If precise simulation can be achieved, high-precision digital twins can be realized.

The model proposed in this paper can only preliminarily implement the deformation-seepage digital twin for cross-sections of high earth-rock dams. Issues such as how to achieve a three-dimensional high-precision digital twin, the influence of the number of sensor monitoring points on digital twin accuracy, how to reduce the limitations of numerical simulations on digital twin accuracy, and computational models for slope stability analysis based on twin data, still need to be addressed in future research.

CRediT authorship contribution statement

Jichen Tian: Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Ruijie Yu:** Writing – original draft, Visualization, Methodology, Data curation. **Jiankang Chen:** Writing – original draft, Supervision, Resources, Funding acquisition. **Chen Chen:** Writing – review & editing, Writing – original draft, Resources, Investigation, Funding acquisition. **Yanling Li:** Writing – original draft, Resources, Investigation, Funding acquisition. **Xinjian Sun:** Writing – original draft, Supervision, Resources. **Huibao Huang:** Writing – original draft, Data curation.

Declaration of competing interest

The author(s) declared no potential conflicts of interest concerning the research, authorship, and/or publication of this article.

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Data availability

Data will be made available on request.

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